12th HACKERS KA JUGAAD

Haloalkanes & Haloarenes

Introduction – Haloalkane (alkyl halide)

halogen

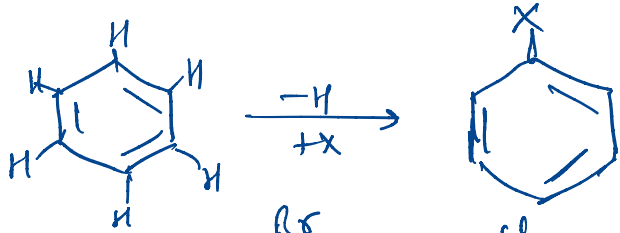
ex - CH_3CH_2Br

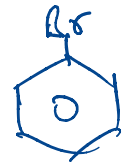
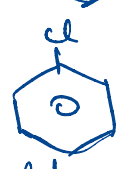
$CH_2 - CH_2$
$Br \quad Br$

they are derivative of aliphatic hydrocarbon

$CH_4 \xrightarrow[+X]{-H} CH_3X$, where X is halogen atom

Haloarenes (aryl halide) → aromatic hydrocarbon



ex  ,  , etc.

no. of bond + lone pair

hybridisation

$\sigma + 4p$

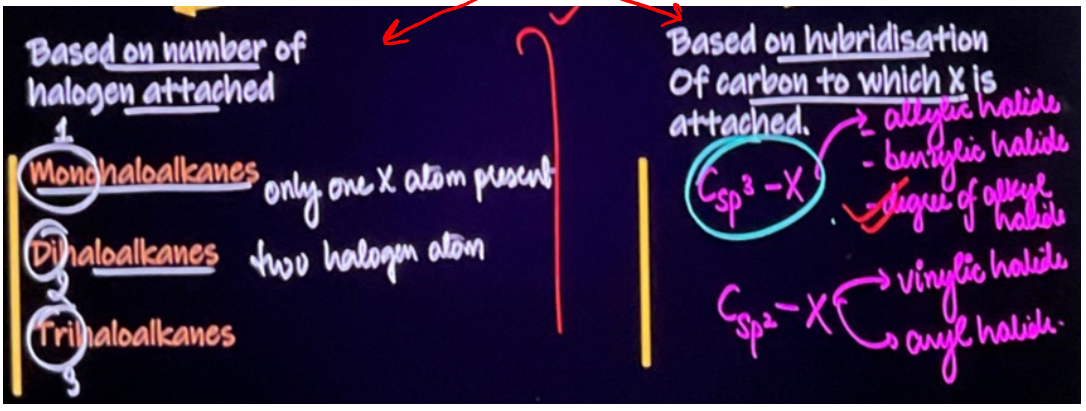
4 sp^3 — ← σ

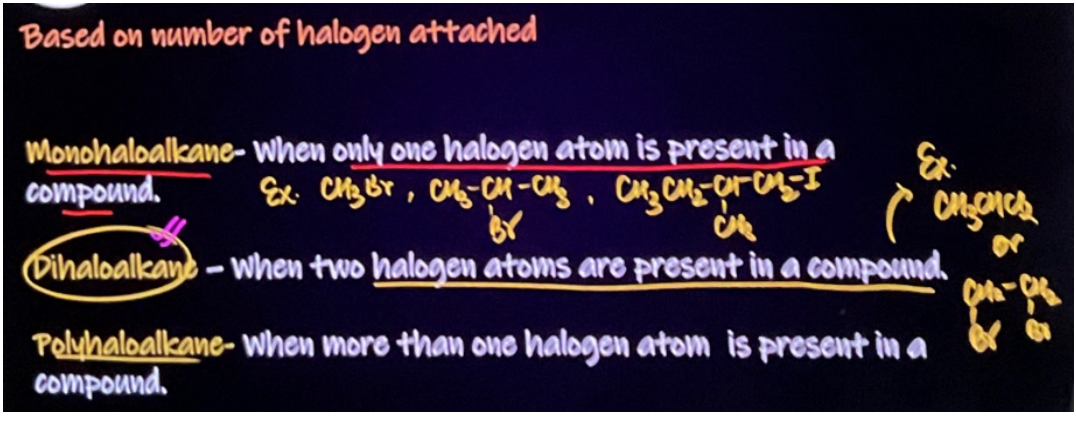
3 sp^2 = π σ

2 sp ≡ π π σ

The replacement of hydrogen atom(s) in an aliphatic or aromatic hydrocarbon by halogen atom(s) results in the formation of alkyl halide (haloalkane) and aryl halide (haloarene), respectively. Haloalkanes contain halogen atom(s) attached to the sp^3 hybridised carbon atom of an alkyl group whereas haloarenes contain halogen atom(s) attached to sp^2 hybridised carbon atom(s) of an aryl group. Many halogen containing organic compounds occur in nature and some of these are clinically useful. These classes of compounds

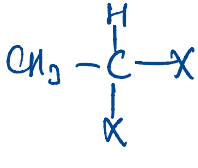
Classification of Haloalkanes





Dihaloalkane

gem-dihalide
or
geminal dihalide
or
gem-dihaloalkane



vic. dihalide
or
vicinal dihaloalkane



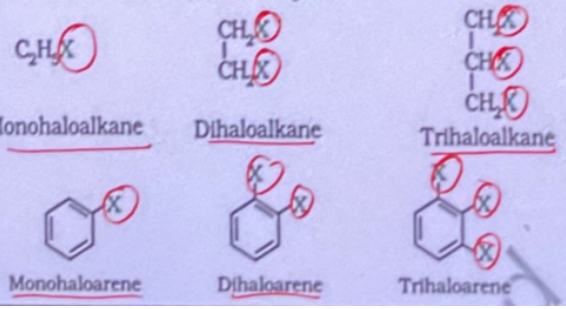
NCERT

6.1 Classification

Haloalkanes and haloarenes may be classified as follows:

6.1.1 On the Basis of Number of Halogen Atoms

These may be classified as mono, di, or polyhalogen (tri-, tetra-, etc.) compounds depending on whether they contain one, two or more halogen atoms in their structures. For example,



↳ Based on hybridisation of carbon atom to which halogen is attached.

$C_{sp^3} - X$

alkyl halide - (degree)

Primary alkyl halide (1°)

$CH_3 - CH_2 - X$

1°C

Secondary alkyl halide (2°)

$CH_3 - \underset{CH_3}{\overset{}{CH}} - X$

Tertiary alkyl halide (3°)

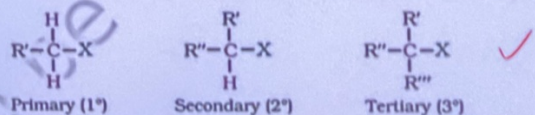
$CH_3 - \underset{CH_3}{\overset{CH_3}{C}} - X$

NCERT

$C_nH_{2n+1}X$ C_nH_{2n+2}

(a) Alkyl halides or haloalkanes (R—X)

In alkyl halides, the halogen atom is bonded to an alkyl group (R). They form a homologous series represented by $C_nH_{2n+1}X$. They are further classified as primary, secondary or tertiary according to the nature of carbon to which halogen is attached. If halogen is attached to a primary carbon atom in an alkyl halide, the alkyl halide is called primary alkyl halide or 1° alkyl halide. Similarly, if halogen is attached to secondary or tertiary carbon atom, the alkyl halide is called secondary alkyl halide (2°) and tertiary (3°) alkyl halide, respectively.



↳ Based on hybridization of carbon atom to which halogen is attached.

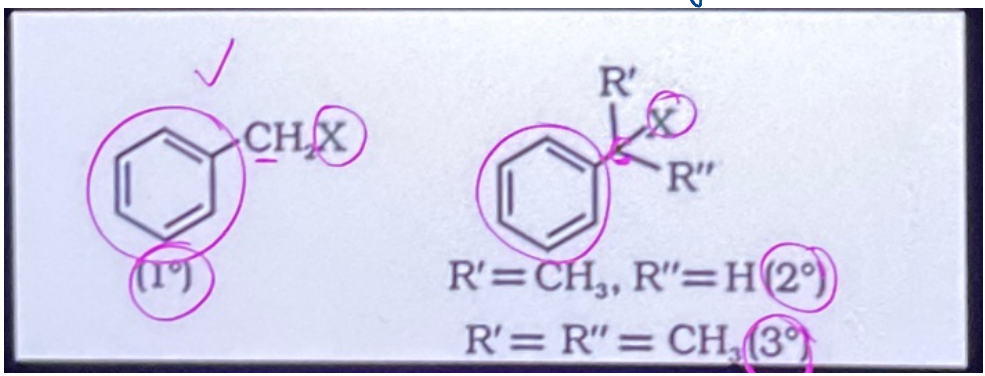
$C_{sp^3} - X$ $\quad$ $C=C-C(sp^3)-X$ (allylic carbon)

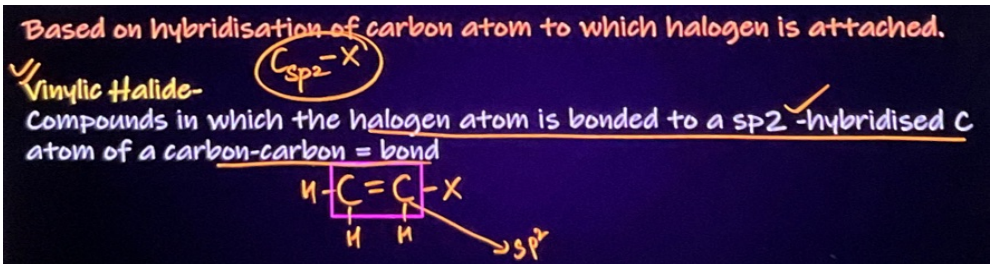
allylic alkyl halide –
Compounds in which the halogen atom is bonded to an sp^3-hybridised C atom further attached to carbon–carbon double bond (C=C)

↳ based on hybridisation of carbon atom to which halogen is attached.

$C_{sp^2} - X$

Benzylic halide – compounds in which the halogen atom is bonded to an sp^3 hybridised carbon atom attached to an aromatic ring.



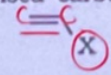


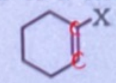
Compounds Containing sp^2 C—X Bond

This class includes:

(a) Vinylic halides

These are the compounds in which the halogen atom is bonded to a sp^2-hybridised carbon atom of a carbon-carbon double bond (C = C).

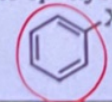


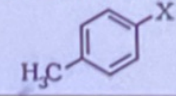


Haloarenes

(b) Aryl halides

These are the compounds in which the halogen atom is directly bonded to the sp^2-hybridised carbon atom of an aromatic ring.





Nomenclature

2° prefix + 1° prefix + word root + 1° suffix + 2° suffix.

- 2° prefix → Junior most fun^n group / side chain / extra group
  - Cl, Br, I
  - chloro, bromo, iodo
- 1° prefix → used for cyclic compounds
  - cyclic → cyclo
- word root → total no. of carbon atoms present in main chain
  - C_1 = meth
  - C_2 = eth
  - C_3 = prop
  - C_4 = but
  - C_5 = pent
  - C_6 = hex
  - C_7 = hept
  - C_8 = oct
  - C_9 = non
  - C_{10} = dec
  - C_{11} = undec
  - C_{12} = dodec.
- 1° suffix
  - C–C → ane
  - C=C → ene
  - C≡C → yne
- 2° suffix → used for senior-most functional group.

6.1 Write structures of the following compounds:

(i) 2-Chloro-3-methylpentane

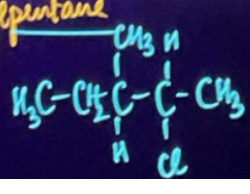
(ii) 1-Chloro-4-ethylcyclohexane

(iii) 4-tert. Butyl-3-Iodoheptane

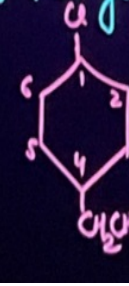
(iv) 1,4-Dibromobut-2-ene

(v) 1-Bromo-4-sec. butyl-2-methylbenzene.

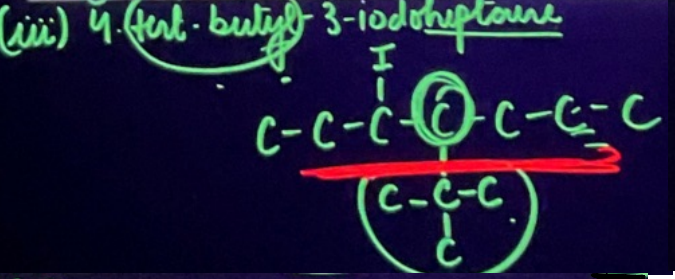
(i) 2-Chloro-3-methylpentane



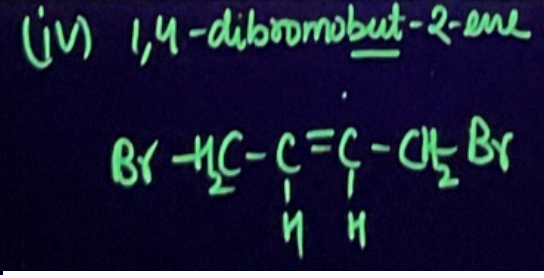
(ii) 1-chloro-4-ethylcyclohexane



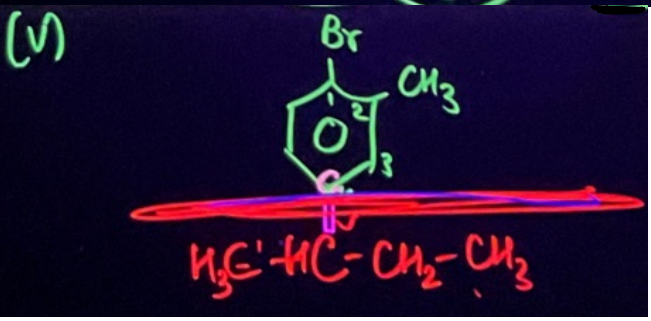
(iii) 4-(tert-butyl)-3-iodoheptane



(iv) 1,4-dibromobut-2-ene



(v)

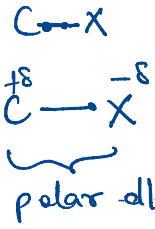


Nature of C-X bond

C-X = EN difference

CH_3-CH_2-X

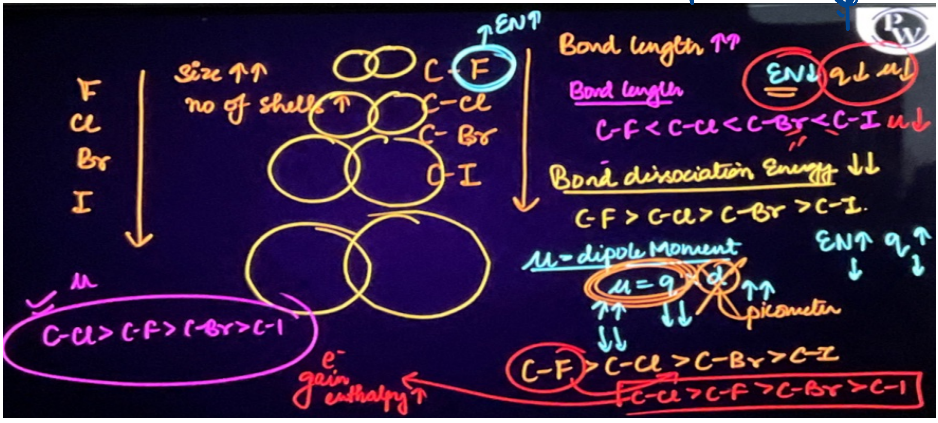




polar dlt difference in EN.

EN → tendancy to attract shared pair of e^- towards itself.

In Monohalo compounds →



Method of preparation of Haloalkanes

1) From alcohol (R-OH)

By H-X
By PCl_3
By PCl_5
By $SOCl_2$.

↳ By H-X hydrogenhalide

$$R-OH \xrightarrow[ZnCl_2]{HX} R-X + H_2O$$

↳ dehydrating agent
↳ removes H_2O.

$CH_3CH_2-OH \xrightarrow[ZnCl_2]{HCl} CH_3CH_2Cl$
1° alcohol

$CH_3-CH(CH_3)-OH \xrightarrow[ZnCl_2]{HCl} CH_3-CH(CH_3)-Cl$
2° alcohol

$(CH_3)_3C-OH \xrightarrow[ZnCl_2]{HCl} (CH_3)_3C-Cl$
3° alcohol

If excess
Red P + $Cl_2 \rightarrow PCl_5$.

↳ By PCl_5

$$R-OH + PCl_5 \longrightarrow R-Cl + POCl_3 + HCl$$

Red P + Cl_2
If Cl_2 limits
$Cl_2 \rightarrow PCl_3$

- By the action of dry HCl on Ethyl alcohol in the presence of anhydrous $ZnCl_2$

$$C_2H_5OH + HCl \xrightarrow[\Delta]{ZnCl_2} C_2H_5Cl + H_2O$$

- Anhydrous $ZnCl_2$ acts as dehydrating agent.
- The process of using HCl gas and anhydrous $ZnCl_2$ is called Grove's process.
- Order of reactivity of alcohols with a given haloacid HX is $3^\circ > 2^\circ > 1^\circ$
- Order of reactivity among acid:

  HX : HI > HBr > HCl > HF

over 1° & 2° alcohol.

↳ By PCl_3

$$3\,R\text{-}OH + PCl_3 \longrightarrow 3R\text{-}Cl + H_3PO_3$$

↳ By $SOCl_2$ (Thionylchloride)

Darzen's process

$$R\text{-}OH + SOCl_2 \xrightarrow{\text{pyridine}} R\text{-}Cl + SO_2\uparrow + HCl\uparrow$$

R-Cl → Haloalkane (pure form)

2.) From hydrocarbon.

from alkanes --- free radical halogenation

Halogenation = add^n of X_2.

$$CH_4 + Cl_2 \xrightarrow[\Delta]{h\nu} CH_3Cl + HCl$$

$$CH_3Cl \xrightarrow{Cl_2} CH_2Cl_2 + HCl \xrightarrow{Cl_2} CHCl_3 + HCl$$

$$CHCl_3 \xrightarrow{Cl_2} CCl_4 + HCl$$

Mechanism of Free Radical Substitution

1) Chain Initiation step
2) Chain propagation
3) Chain Termination

CH_4

$CH_3Cl + Cl-Cl \rightarrow CH_2Cl_2 + \dot{Cl}$

$CH_4 + \dot{Cl} \rightarrow HCl + \dot{C}H_3$

Cl_2

Jab Jab C Khatre Mein hoga, Cl_2 Molecule आयेगा।

1) Chain Initiation step

$Cl-Cl \rightarrow \dot{Cl} + \dot{Cl}$

2) Chain propagation step

$CH_3-H + \dot{Cl} \rightarrow HCl + \dot{C}H_3$

$\dot{C}H_3 + Cl-Cl \rightarrow CH_3Cl + \dot{Cl}$

$CH_2(Cl)-H + \dot{Cl} \rightarrow HCl + \dot{C}H_2Cl$

3.) chain termination step.

$\dot{Cl} + \dot{Cl} \rightarrow Cl_2$

$\dot{C}H_3 + \dot{Cl} \rightarrow CH_3Cl$

$\dot{C}H_3 + \dot{C}H_3 \rightarrow CH_3-CH_3$

From alkenes ----- Addition of hydrogen halide

HX

$CH_2=CH_2 + H^+X^- \rightarrow CH_3CH_2X$

Symmetrical alkene

$CH_3-CH=CH_2 + H^+X^- \rightarrow CH_3-CH(X)-CH_3 + CH_3-CH_2-CH_2-X$

Unsymmetrical alkene

(Markovnikov Rule) → -ve part will go to that double bond carbon atom which has less hydrogen atoms

$CH_3-CH=CH_2 + H^+Cl^- \rightarrow CH_3-CH(Cl)-CH_3$ (Major) + $CH_3-CH_2-CH_2-Cl$ (Minor).

$CH_3-CH=CH_2 + H^+Br^- \rightarrow CH_3-CH(Br)-CH_3$

(Markovnikov Rule)

* HBr + peroxide = Anti-Markovnikov Rule = -ve part will go that double Bond C atom which has more no. of H atoms

$CH_3-CH=CH_2 + H^+Br^- \xrightarrow{\text{peroxide}} CH_3-CH_2-CH_2-Br$ (Major) + $CH_3-CH(Br)-CH_3$ (Minor).

from alkenes – add" of halogens Br

$CH_2=CH_2 \xrightarrow[CCl_4]{Br_2} CH_2(Br)-CH_2(Br)$

3.) Halogen exchange reaction –

Swarts rea^n → R–F

Finkelstein rea^n → R–I

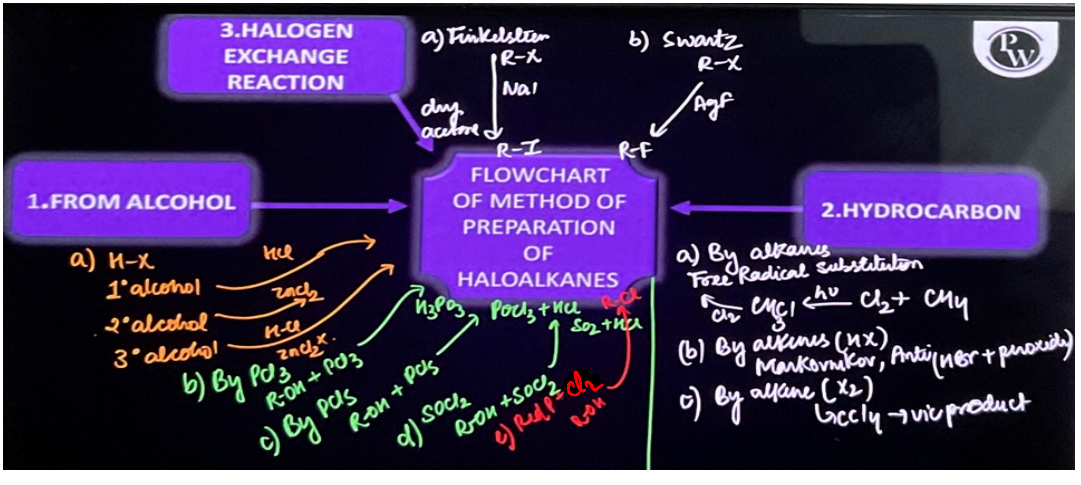
Halogen Exchange Reations

Finkelstein Reation

- Alkyl iodides are prepared by the reaction of alkyl chlorides or bromides with NaI in dry acetone.

$R-X + NaI \xrightarrow{\text{Dry Acetone}} R-I + NaX$ (X = Cl or Br)

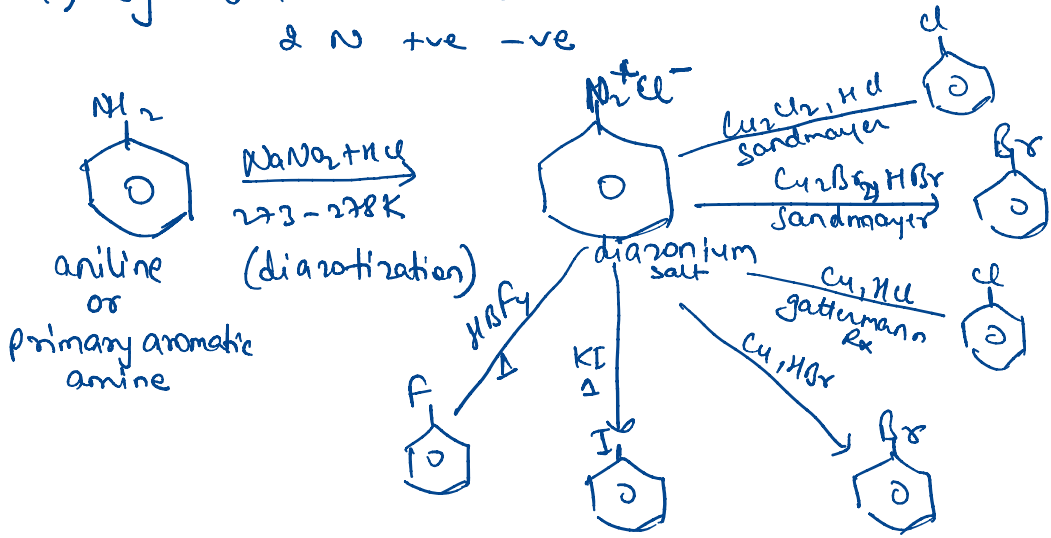
- NaCl or NaBr formed is precipitated in dry acetone. It facilitates the forward reaction according to Le-chatelier's principle.
- The reaction follows S_N2 mechanism.
- **SWARTS REACTION -**
- Alkyl fluorides are prepared by heating alkyl chloride on bromide in presence of metallic fluoride such as AgF, Hg_2F_2, CoF_2 or SbF_3
- $R-Br + AgF \rightarrow R-F + AgBr$



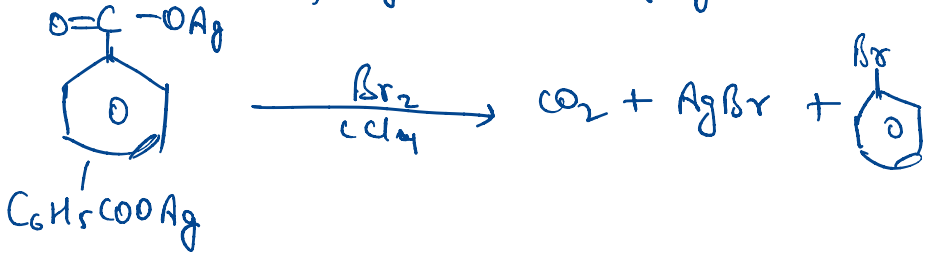
Methods of Preparation of Haloarenes

1.) By diazonium salt

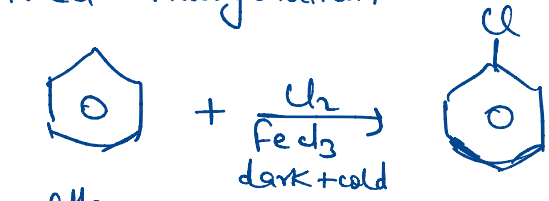
2 N +ve −ve

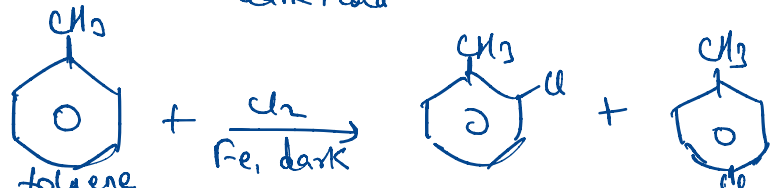


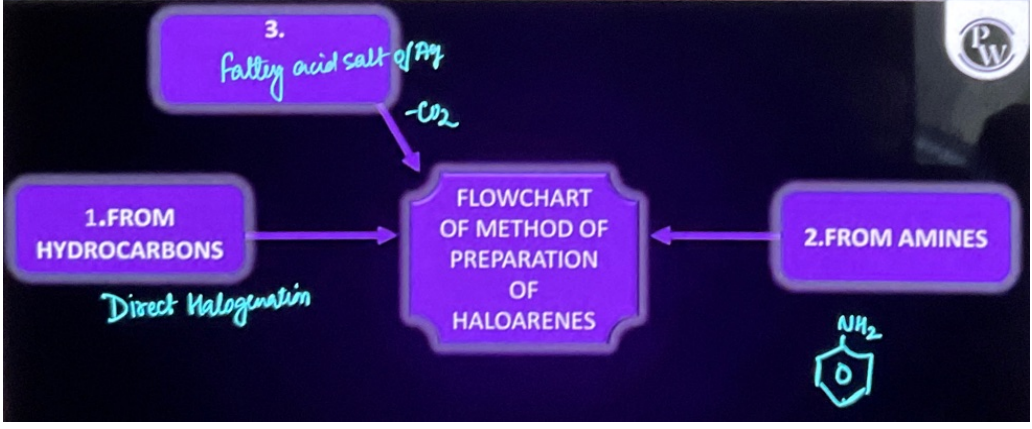
2.) Hunsdicker — fatty acid salt of Ag



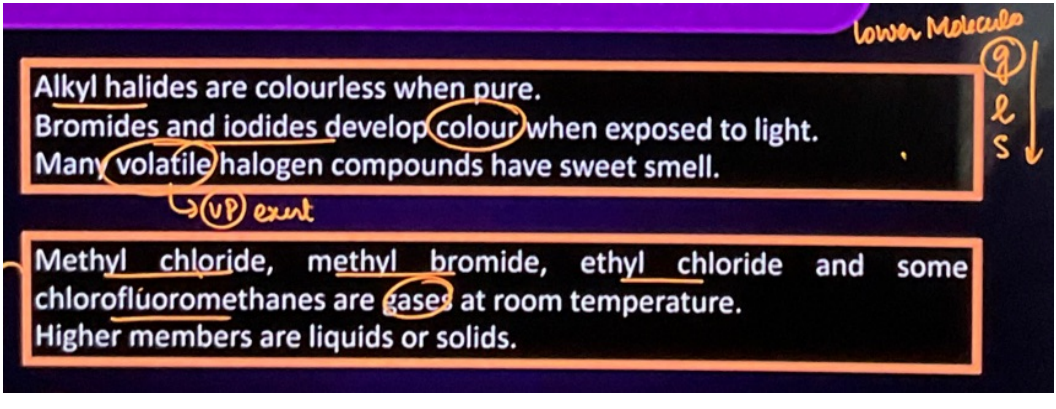
3.) Direct halogenation

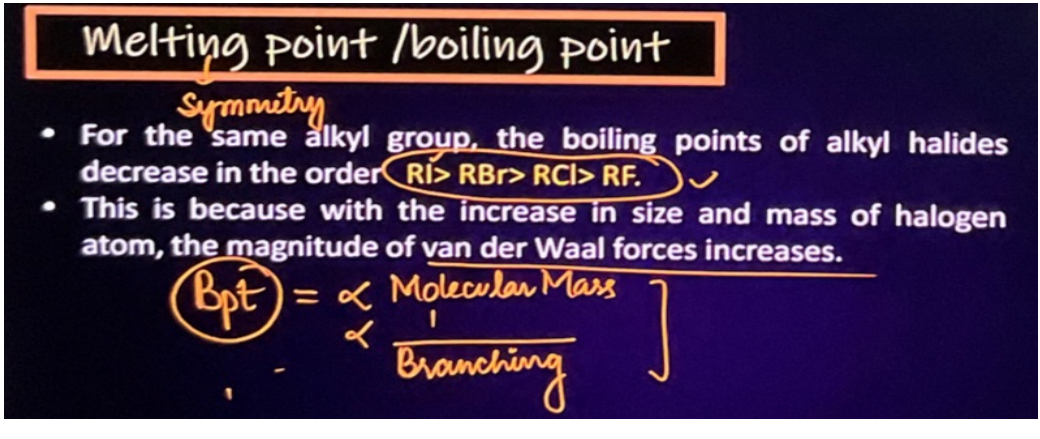






Physical properties of Haloalkanes





Density

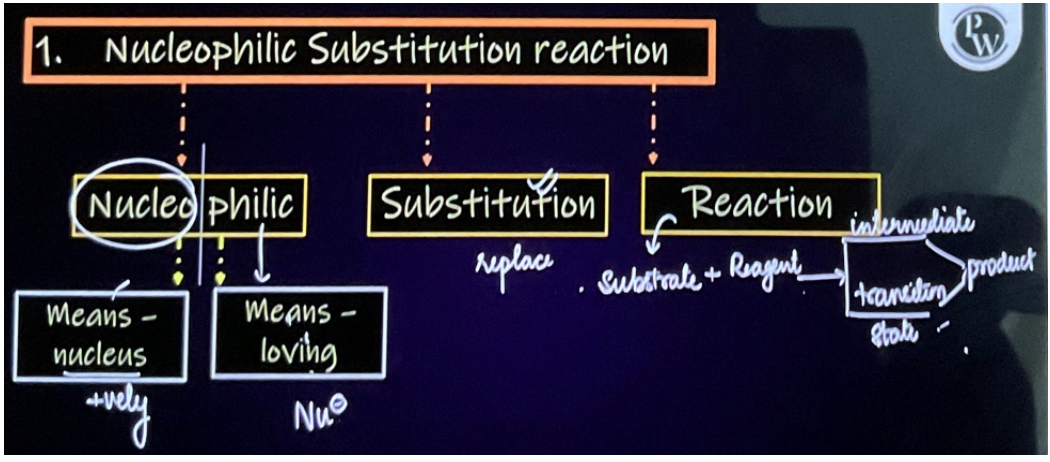
Bromo, Iodo and polychloro derivatives of hydrocarbons are heavier than water. The density increases with increase in number of carbon atoms, halogen atoms and atomic mass of halogen atoms. $d = m/v$

Solubility

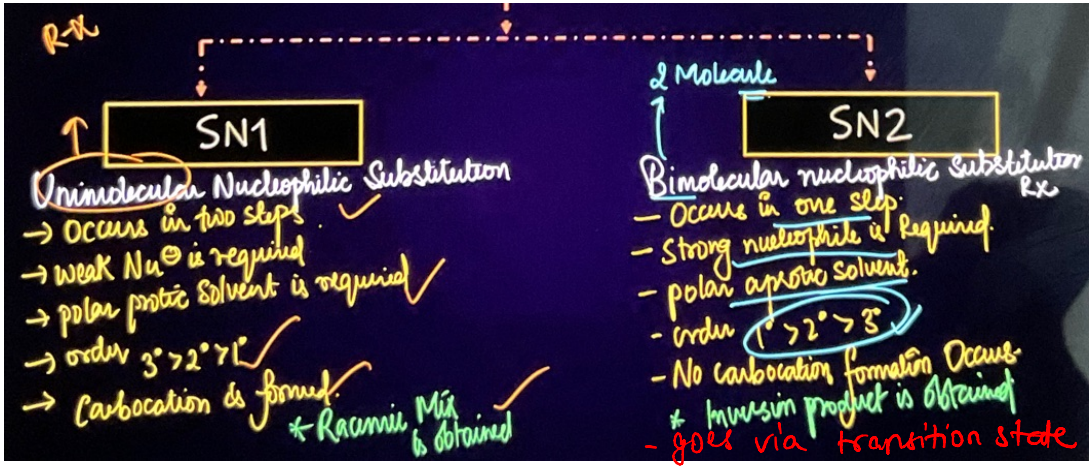
The Haloalkanes are very slightly soluble in water. In order to dissolve halo alkane in water. Energy is required to overcome the attractions between the haloalkane molecules and break the hydrogen bonds between water molecules. less energy is released when new attractions are set up a betweeen haloalkane and the water molecules as these are not as strong as the original hydrogen bonds in water.

Solubility ↓↓ C↑↑ hydrophobic ↑ H_2O hate

Chemical properties of Haloalkane



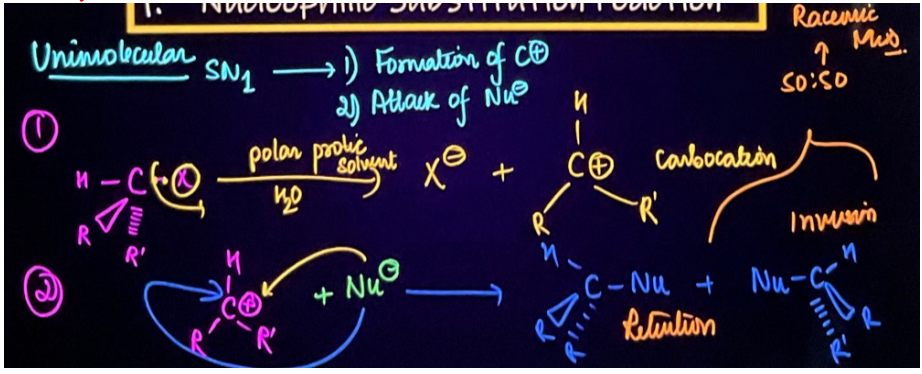
Types of nucleophilic Substitution Reaction

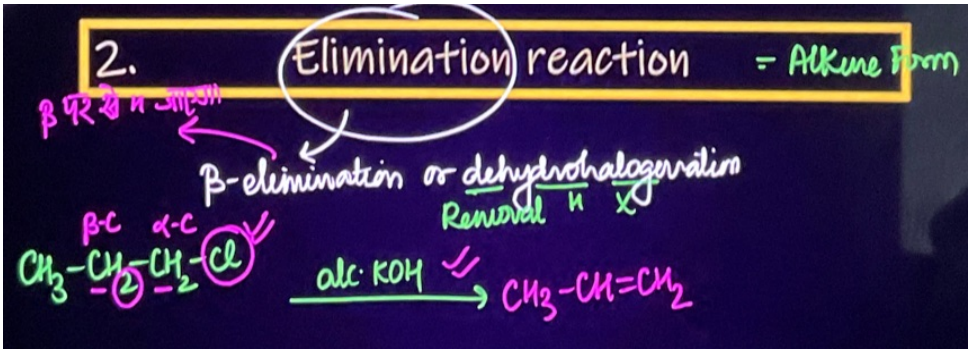


SN2 – single step



SN1 – two step.





$$CH_3-\underset{\beta\text{-C}}{CH}(Cl)-\underset{\alpha\text{-C}}{CH_2}-CH_3 \xrightarrow{\text{alc. KOH}} \underset{\alpha\text{-C}}{CH_3}-CH=CH-\underset{\alpha\text{-C}}{CH_2}-CH_3 + CH_2=\underset{\alpha\text{-C}}{CH}-CH_2-CH_3$$

α-H = 6 (Saytreff product) ; α-H = 2 (Hoffmann product)

Saytreff rule – more the no. of α-H atoms
Major = Saytreff product
minor = Hoffmann product.

3.) **Reaction with Metals**

$$R-X + Mg \xrightarrow{\text{dry ether}} RMgX$$

Grignard Reagent
↳ organometallic compound

$$R\overset{\delta^-}{-}Mg^{\delta^+}X + H_2O \longrightarrow R-H + Mg(OH)X$$

(H^+ OH^-)

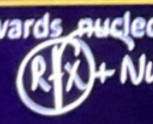
hydrocarbon.

4.) **Wurtz Rx**

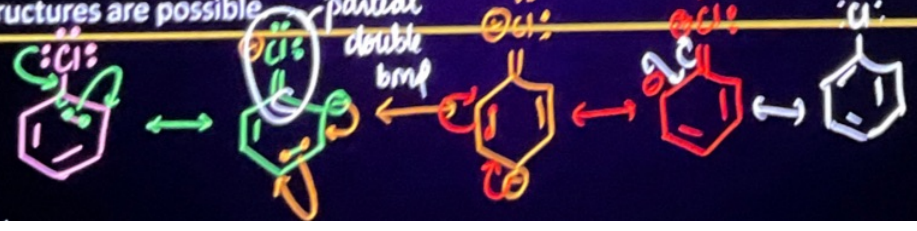
$$2R-X + 2Na \xrightarrow[\text{ether}]{\text{dry}} 2NaX + R-R$$

Chemical properties of Haloarenes

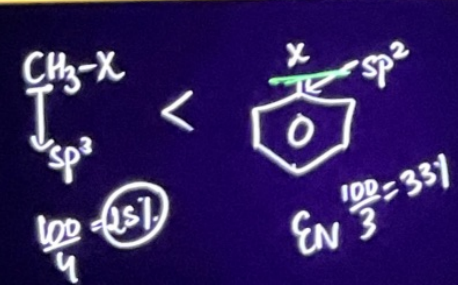
Aryl halides are extremely less reactive towards nucleophilic substitution reactions due to the following reasons:



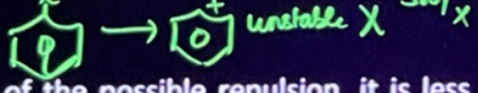
(i) Resonance effect : In haloarenes, the electron pairs on halogen atom are in conjugation with p-electrons of the ring and the following resonating structures are possible.



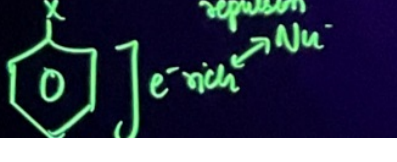
(ii) Difference in hybridisation of carbon atom in C—X bond:
In haloalkane, the carbon atom attached to halogen is sp3 hybridised while in case of haloarene, the carbon atom attached to halogen is sp2 - hybridised.



iii) Instability of phenyl cation: In case of haloarenes, the phenyl cation formed as a result of self-ionisation will not be stabilised by resonance and therefore, SN 1 mechanism is ruled out.



(iv) Because of the possible repulsion, it is less likely for the electron rich nucleophile to approach electron rich arenes.

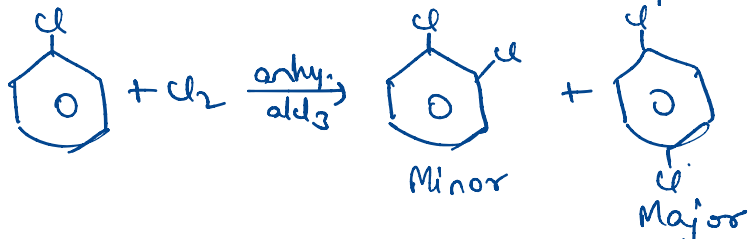


Electrophilic Substitution Rx.

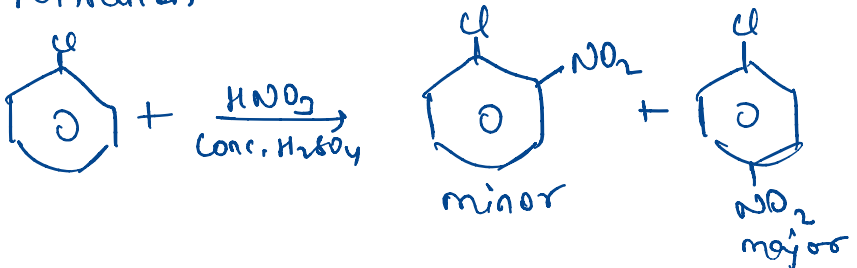
$Cl-Cl \rightarrow Cl^{\oplus} + Cl^{\ominus}$ — electrophile.

① Halogenation

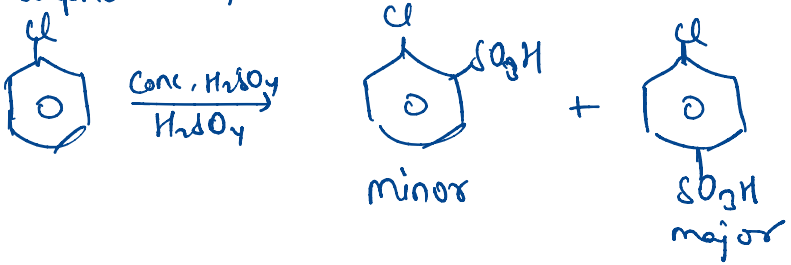
e⁻ loving



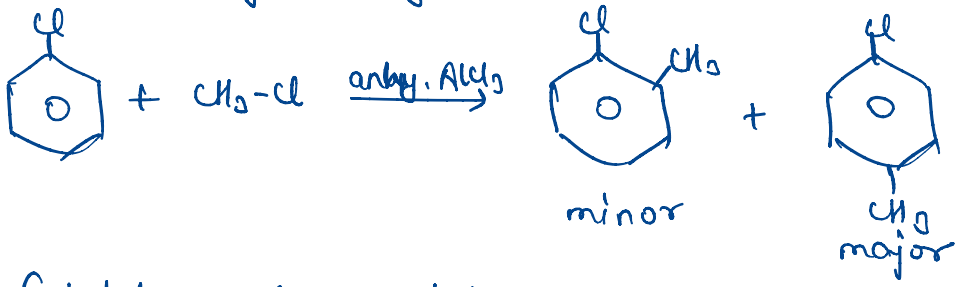
② Nitration



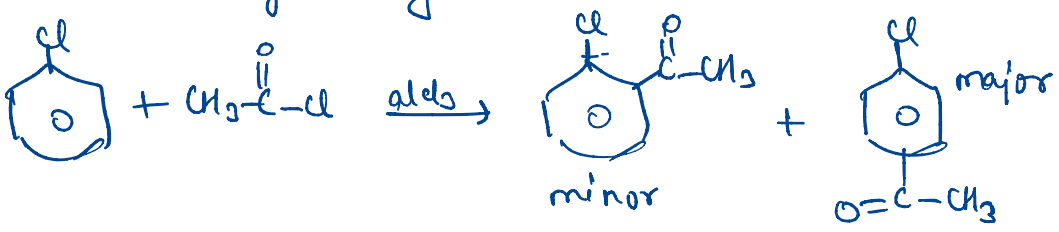
③ Sulphonation



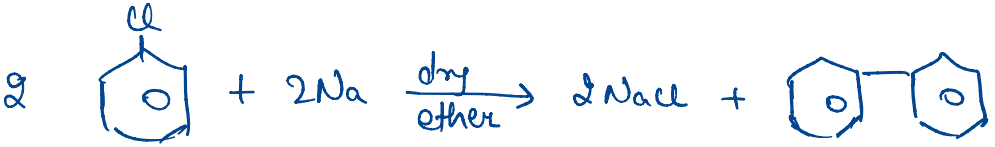
④ Friedel craft alkylation



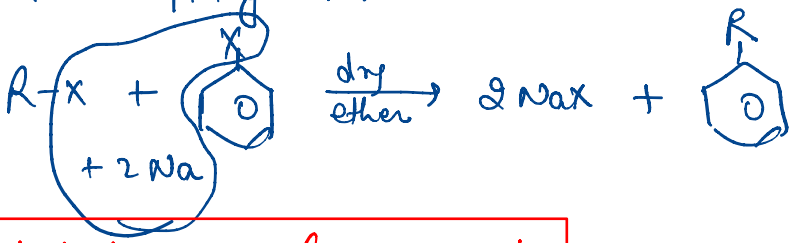
⑤ Friedel craft acylation.



→ Fittig Rx



→ Wurtz - fittig Rx



Polyhalogen Compounds

CH_2Cl_2 Dichloromethane (Methylene chloride)-

Used as a solvent as a paint remover, as a propellant in aerosols, and as a process solvent in the manufacture of drugs.
Also used as a metal cleaning and finishing solvent.
Methylene chloride harms the human central nervous system. Exposure to lower levels of methylene chloride in air can lead to slightly impaired hearing and vision.
Higher levels of methylene chloride in air cause dizziness, nausea, tingling and numbness in the fingers and toes. In humans, direct skin contact with methylene chloride causes intense burning and mild redness of the skin. Direct contact with the eyes can burn the cornea

Trichloromethane (Chloroform)- $CHCl_3$

The major use of chloroform today is in the production of the freon refrigerant R-22. It was once used as a general anaesthetic in surgery but has been replaced by less toxic, safer anaesthetics, such as ether.

Chronic chloroform exposure may cause damage to the liver (where chloroform is metabolised to phosgene) and to the kidneys, and some people develop sores when the skin is immersed in chloroform.

$CHCl_3 + O_2 \longrightarrow$ Phosgene = Carbonyl chloride $COCl_2$

Chloroform is slowly oxidised by air in the presence of light to an extremely poisonous gas, carbonyl chloride, also known as phosgene. It is therefore stored in closed dark coloured bottles completely filled so that air is kept out.

Triiodomethane (iodoform)-

It was used earlier as an antiseptic but the antiseptic properties are due to the liberation of free iodine and not due to iodoform itself.
Due to its objectionable smell, it has been replaced by other formulations containing iodine.

FREONS-

The chlorofluorocarbon compounds of methane and ethane are collectively known as freons. They are extremely stable, unreactive, non-toxic, noncorrosive and easily liquefiable gases. Freon 12 (CCl_2F_2) is one of the most common freons in industrial use. It is manufactured from tetrachloromethane by Swarts reaction.

Tetrachloromethane (carbontetrachloride)-

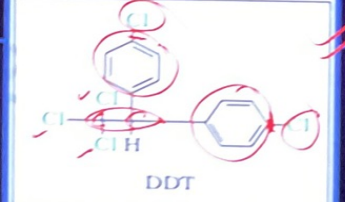
CCl_4.

It is produced in large quantities for use in the manufacture of refrigerants and propellants for aerosol cans. It is also used as feedstock in the synthesis of chlorofluorocarbons and other chemicals, pharmaceutical manufacturing, and general solvent use.

The most common effects are dizziness, light headedness, nausea and vomiting, which can cause permanent damage to nerve cells. In severe cases, these effects can lead rapidly to stupor, coma, unconsciousness or death. Exposure to CCl_4 can make the heart beat irregularly or stop. The chemical may irritate the eyes on contact. When carbon tetrachloride is released into the air, it rises to the atmosphere and depletes the ozone layer.

P,P'-Dichlorodiphenyltrichloroethane(DDT)-

DDT is not metabolised very rapidly by animals; instead, it is deposited and stored in the fatty tissues. If ingestion continues at a steady rate, DDT builds up within the animal over time. The use of DDT was banned in the United States in 1973, although it is still in use in some other parts of the world.



DDT

DDT